EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Discovery of subatomic particles (Discharge Tube Experiment; Cathode rays) charge to mass ratio(e/m)

1. The specific charge is maximum for

(A) $Na^{\oplus}$ (B) $Al^{3\oplus}$ (C) $H^{\oplus}$ (D) $Mg^{2\oplus}$

CAS094

2. The mass of 1 mole of electron is :–

(A) 0.55 mg (B) 9.1×10^{-28} g (C) 9.1×10^{-27} g (D) 1.008 mg

CAS095

Anode Rays; Discovery of proton, neutron

3. Anode rays are made up of:

(A) only protons (B) only nucleus of atom

(C) positive residue of atoms (D) only electrons

CAS097

4. When Lithium $\left({}^{7}_{3}\text{Li}\right)$ vapours were filled in discharge tube for anode ray experiment, the anode rays were found to contain only $Li^{\oplus}$ ions. Thus the anode ray particle contains :

(A) 1 proton only (B) 3 proton and 4 neutrons only

(C) 3 proton, 4 neutrons and 2 electrons (D) 3 proton, 3 neutrons and 3 electrons

CAS098

Rankers Academy JEE

5. The specific charge of a proton is 9.6×10^{7}C/kg then for an α particles it will be :

(A) 2.4×10^{7} C/kg (B) 4.8×10^{7} C/kg

(C) 19.2×10^{7} C/kg (D) 38.4×10^{7} C/kg

CAS099

6. The ratio of the (e/m) values of a proton & a deuterium is :

(A) 1 : 1 (B) 1 : 2 (C) 1 : 0 (D) 2 : 1

CAS101

Atomic models (Thomson Model of Atom , Rutherford Nuclear Model of Atom)

7. α–particles are projected towards the nucleus of following metals, with the same kinetic energy. The distance of closest approach will be minimum for which metal?

(A) Cu (B) Ag (C) Au (D) Ca

CAS104

8. Which of the following is not a conclusion of Rutherford's atomic model –

(A) Most of the part inside an atom is empty

(B) Almost all mass of an atom is concentrated in the nucleus.

(C) The size of nucleus is very small in comparison to the size of atom

(D) Electron revolves around the nucleus in definite orbits.

CAS105

9. In the different experiments, α–particles, proton, deuteron and neutron are projected towards gold nucleus with the same kinetic energy. The distance of closest approach will be minimum for

(A) α–particle (B) proton (C) deuteron (D) neutron

CAS106

For More Join: @IITJEE_Advanced

10. Rutherford's experiment which established the nuclear model of the atom, used a beam of :–
(A) β–particles, which impinged on metal foil and got absorbed
(B) γ–rays, which impinged on a metal foil and got ejected electrons
(C) Helium atoms, which impinged on a metal foil and got scattered
(D) Helium nuclei which impinged on a metal foil and got scattered

CAS107

11. The nucleus of an atom can be assumed to be spherical. The radius of the nucleus of mass number A is given by $1.25 \times 10^{-13} \times A^{1/3}$ cm. Radius of atom is one Å. If the mass number is 64, then the fraction of the atomic volume that is occupied by the nucleus is
(A) 1.0×10^{-3} (B) 5.0×10^{-5} (C) 2.5×10^{-2} (D) 1.25×10^{-13}

CAS108

Atomic Number and Mass Number

12. Atoms consist of electrons, protons and neutrons. If the mass attributed to neutron were halved and that attributed to the electrons were doubled, the atomic mass of ${}_6C^{12}$ would be approximately :–
(A) same (B) doubled (C) halved (D) reduced by 25%

CAS100

13. e/m ratio of a particle of charge 2 unit and mass 4 amu is
(A) 4.8×10^7 C/kg (B) 0.5 C/kg (C) 4.8×10^4 C/kg (D) 8×10^{-20} C/kg

CAS187

Developments leading to the Bohr's model of atom; Maxwell's Electromagnetic Wave Theory; Wave characteristics such as wavelength, frequency, time period, wave number; Electromagnetic Spectrum

Bankers Academy JEE

14. The wavelength of 5000 Å lies in.
(A) Ultraviolet region (B) Infrared region
(C) Visible region (D) Can't say

CAS110

15. Calculate the wavelength of an ultraviolet wave, if it's frequency is 12×10^{16} cycles/sec & $C = 3 \times 10^8$ m/sec:–
(A) 2.5×10^{-10} m (B) 2.5×10^{-8} m
(C) 2.5×10^{-9} m (D) 2.5×10^{-12} m

CAS111

16. Calculate the wave number of a radiation having wavelength 600 nm:–
(A) 1.67×10^8 m^{-1} (B) 1.67×10^{10} m^{-1} (C) 1.67×10^{12} m^{-1} (D) 1.67×10^6 m^{-1}

CAS112

17. Which of the following electromagnetic radiation have greater frequency ?
(A) X – rays (B) Ultraviolet rays (C) Radio waves (D) Visible rays

CAS189

18. A radio station is emitting the radiations of frequency 2×10^4 Hz. If its frequency is doubled,
(A) wavelength will be doubled (B) energy per quanta will be doubled
(C) wave number will be halved (D) all of these

CAS190

For More Join: @IITJEE_Advanced

Particle Nature of Electromagnetic Radiation: Planck's Quantum Theory

19. Two electromagnetic radiations have wave numbers in the ratio 2 : 3. Their energies per quanta will be in the ratio

(A) 3 : 2 (B) 9 : 4 (C) 4 : 9 (D) 2 : 3

CAS114

20. A photon of 400 nm is absorbed by a gas molecule and then the molecule re–emits two photons. One re–emitted photon has wavelength 500 nm. Assuming that there is no change in the energy of molecule, the wavelength of second re–emitted photon is

(A) 100 nm (B) 2000 nm (C) –100 nm (D) 900 nm

CAS115

21. A green bulb and a red bulb are emitting the radiations with equal power. The correct relation between numbers of photons emitted by the bulbs per second is

(A) $n_g = n_r$ (B) $n_g < n_r$ (C) $n_g > n_r$ (D) unpredictable

CAS116

22. The energy of a photon of light is _____ proportional to its frequency and _____ proportional to its wavelength.

(A) directly, directly (B) inversely, inversely

(C) inversely, directly (D) directly, inversely

CAS117

23. If power of a bulb is 80 watt and wavelength of emitted radiation is 5000 Å, find number of photons emitted per second :–

(A) 2×10^{10} (B) 2×10^{30} (C) 12×10^{10} (D) None of these

CAS118

Rankers Academy JEE

24. The ratio of the energy of a photon of 2000 Å wavelength radiation to that of 4000 Å radiation is

(A) 1 / 4 (B) 4 (C) 1 / 2 (D) 2

CAS191

Photoelectric Effect

25. Electromagnetic radiations having wavelength 310 Å are subjected to a metal sheet having work function of 12.8 eV. What will be the velocity of photoelectrons with maximum kinetic energy.

(A) 0, no emission will occur (B) 2.18×10^6 m/s

(C) $2.18 \sqrt{2} \times 10^6$ m/s (D) 8.72×10^6 m/s

CAS119

26. Radiation of wavelength 155 nm was irradiated on Li (work function = 5 eV) plate. The stopping potential (in V) is _____.

(A) 3 V (B) 8 V (C) 9 V (D) 5 V

CAS122

27. The threshold wavelength for ejection of electrons from a metal is 330 nm. The work function for the photoelectric emission from the metal is ($h = 6.6 \times 10^{-34}$ J–s)

(A) 1.2×10^{-18}J (B) 6.0×10^{-19} J (C) 1.2×10^{-20} J (D) 6.0×10^{-12} J

CAS193

28. Radiations of frequency, ν, are incident on a photosensitive metal. The maximum kinetic energy of photoelectrons is 'E'. When the frequency of the incident radiations is doubled, what is the maximum kinetic energy of the photoelectrons ?

(A) 2E (B) E/2 (C) E + hv (D) E – hv

CAS194

For More Join: @IITJEE_Advanced

29. The energy required to remove an electron from the surface of sodium metal is 3.3 eV. What is the longest wavelength of radiation with which it can show photoelectric effect?
Use h = 6.6×10^{-34} J.s
(A) 1.75×10^{-7} m (B) 3.75×10^{-5} m (C) 1.75×10^{-5} m (D) 3.75×10^{-7} m

CAS206

Bohr's Model for Hydrogen Atom

30. In which of the following is the radius of the first orbit minimum ?
(A) A Hydrogen atom (B) A tritium atom
(C) Triply ionized beryllium (D) Double ionized helium

CAS123

31. Choose the incorrect relations on the basis of Bohr's theory
(A) Velocity of electron $\propto \frac{1}{n}$ (B) Frequency of revolution $\propto \frac{1}{n^3}$
(C) Radius of orbit $\propto n^2z$ (D) Force on electron $\propto \frac{1}{n^4}$

CAS125

32. The energy of hydrogen atom in its ground state is –13.6 eV. The energy of the level corresponding to n = 5
(A) –0.54 eV (B) –5.40 eV (C) –0.85 eV (D) –2.72 eV

CAS126

33. If the mass of electron is doubled, the radius of first orbit of H–atom become about
(A) 0.529 Å (B) 0.265 Å (C) 1.058 Å (D) 0.32 Å

CAS129

Rankers Academy JEE

34. The speed of electron revolving in the 4th orbit of a hydrogen like atom or ion is 1094 km/s. The atom or ion is
(A) H (B) $He^{\oplus}$ (C) $Li^{2\oplus}$ (D) $Be^{3\oplus}$

CAS130

35. Bohr's model may be applied to
(A) Na^{10+} ion (B) He atom (C) Be^{2+} ion (D) C^{6+} ion

CAS195

36. The ratio of time taken by electron in revolutions round the H–nucleus in 2nd and 3rd orbits, are
(A) 2 : 3 (B) 4 : 8 (C) 8 : 27 (D) 27 : 8

CAS196

37. Which of the following is not a permissible value of angular momentum of electron in H–atom?
(A) $1.5 \frac{h}{\pi}$ (B) $0.5 \frac{h}{\pi}$ (C) $1.25 \frac{h}{\pi}$ (D) all of these

CAS197

38. Angular momentum for P–shell electron :–
(A) $\frac{3h}{\pi}$ (B) Zero (C) $\frac{\sqrt{2}h}{2\pi}$ (D) None

CAS198

39. Angular momentum in 2nd Bohr orbit of H–atom is x. Then find out angular momentum in Ist excited state of Li^{+2} :–
(A) 3x (B) 9x (C) x/2 (D) x

CAS199

For More Join: @IITJEE_Advanced

40. The orbit from which when electron will jump in other orbit, energy may be absorbed but not emitted out, will be

(A) 1st orbit (B) 2nd orbit (C) 7th orbit (D) infinite orbit

CAS200

41. The potential energy of electron revolving in the ground state of H–atom is

(A) –13.6 eV (B) –6.8 eV (C) –27.2 eV (D) Zero

CAS201

42. If the potential energy (PE) of hydrogen electron is –3.02eV then in which of the following excited level is electron present :–

(A) 1st (B) 2nd (C) 3rd (D) 4th

CAS202

43. A hydrogen atom (ionisation energy 13.6 eV) jumps from third excited state to first excited state. The energy of photon emitted in the process is

(A) 1.89 eV (B) 2.55 eV (C) 12.09 eV (D) 12.75 eV

CAS205

44. Which is correct for any H like species :–

(A) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$ (B) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$

(C) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$ (D) $(E_2 - E_1) = 1/4\ (E_3 - E_2) = 1/9\ (E_4 - E_3)$

CAS207

45. A single electron orbits a stationary nucleus of charge +Ze, where Z is a constant. It requires 47.2 eV to excite electron from second Bohr orbit to third Bohr orbit, find the value of Z :–

(A) 1 (B) 3 (C) 5 (D) 4

CAS208

Rankers Academy JEE

46. The energy of H–atom in n^{th} orbit is E_n then energy in n^{th} orbit of singly ionised helium atom will be:

(A) $4E_n$ (B) $E_n/4$ (C) $2E_n$ (D) $E_n/2$

CAS209

47. **Statement–1** : Emitted radiations will fall in visible range when an electron jump from higher level to n = 2 in Li^{+2} ion.

Statement–2 : Balmer series radiations belong to visible range in all H–atoms.

(A) Statement–1 is true, statement–2 is true and statement–2 is correct explanation for statement–1.

(B) Statement–1 is true, statement–2 is true and statement–2 is NOT the correct explanation for statement–1.

(C) Statement–1 is true, statement–2 is false.

(D) Statement–1 is false, statement–2 is true.

CAS218

Explanation of Line Spectrum of Hydrogen

48. The wavelength of third line of the Balmer series for a H–atom is

(A) $\frac{21}{100R}$ (B) $\frac{100}{21R}$ (C) $\frac{21R}{100}$ (D) $\frac{100R}{21}$

CAS131

49. When the electron of a hydrogen atom jumps from n = 4 to n = 1 state, the number of spectral lines emitted is

(A) 15 (B) 6 (C) 3 (D) 4

CAS132

For More Join: @IITJEE_Advanced

50. If wavelength of last line of Balmer series of $He^{\oplus}$ ion is x Å, then calculate maximum wavelength for Lyman series of H–atom :–

(A) x Å (B) $\frac{3}{4}$x Å (C) $\frac{4}{3}$x Å (D) None

CAS136

51. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen

(A) $5 \rightarrow 3$ (B) $5 \rightarrow 2$ (C) $4 \rightarrow 3$ (D) $4 \rightarrow 2$

CAS211

52. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2$ eV. The same energy change is observed in the second Balmer transition of :–

(A) Li^{2+} (B) Li^{+} (C) He^{+} (D) Be^{3+}

CAS212

53. In a sample of H–atoms, electron transits from 6th orbit to 2nd orbit in multi step. Then total spectral lines (without Balmer series) will be :–

(A) 6 (B) 10 (C) 4 (D) 0

CAS213

54. The shortest wavelength of He^{+} ion in Balmer series is x, then longest wavelength in the Paschen series of Li^{+2} is

(A) $\frac{36x}{5}$ (B) $\frac{16x}{7}$ (C) $\frac{9x}{5}$ (D) $\frac{5x}{9}$

Rankers Academy JEE

CAS214

55. The ratio of wave length of photon corresponding to the α–line of Lyman series in H–atom and β–line of Balmer series in He^{+} is

(A) 1 : 1 (B) 1 : 2 (C) 1 : 4 (D) 3 : 16

CAS215

56. The value of $(n_2 + n_1)$ and $\left(n_2^2 - n_1^2\right)$ for He^{+} ion in atomic spectrum are 4 and 8 respectively. The wavelength of emitted photon when electron jump from n_2 to n_1 is

(A) $\frac{32}{9}\ R_H$ (B) $\frac{9}{32}\ R_H$ (C) $\frac{9}{32R_H}$ (D) $\frac{32}{9R_H}$

CAS216

Towards Quantum Mechanical Model of Atom; De Broglie's Hypothesis

57. An electron in a hydrogen atom in its ground state absorbs energy equal to the ionisation energy of Li^{+2}. The wavelength of the emitted electron is:

(A) 3.32×10^{-10} m (B) 1.17 Å (C) 2.32×10^{-9} nm (D) 3.33 pm

CAS217

58. An electron has a kinetic energy of 2.8×10^{-23} J. De–Broglie wavelength will be nearly :–

($m_e = 9.1 \times 10^{-31}$ kg)

(A) 9.28×10^{-24} m (B) 9.28×10^{-7} m

(C) 9.28×10^{-8} m (D) 9.28×10^{-10} m

CAS137

For More Join: @IITJEE_Advanced

59. If the De–Broglie wavelength of the fourth Bohr orbit of hydrogen atom is 4Å, the circumference of the orbit will be :–

(A) 4Å (B) 4 nm (C) 16 Å (D) 16 nm

CAS138

60. What should be the momentum (in gram centimetre per second) of a particle if its De–Broglie wavelength is 1Å and the value of h is 6.6252×10^{-27} erg second ?

(A) 6.6252×10^{-19} g cm/s (B) 6.6252×10^{-21} g cm/s

(C) 6.6252×10^{-24} g cm/s (D) 6.6252×10^{-27} g cm/s

CAS140

61. An electron, a proton and an alpha particle have kinetic energies of 16E, 4E and E respectively. What is the qualitative order of their de Broglie wavelengths?

(A) $\lambda_e > \lambda_p = \lambda_\alpha$ (B) $\lambda_p = \lambda_\alpha > \lambda_e$ (C) $\lambda_p > \lambda_e > \lambda_\alpha$ (D) $\lambda_\alpha < \lambda_e » \lambda_p$

CAS142

62. The wavelength associated with a golf weighing 200g and moving at a speed of 5m/h is of the order

(A) 10^{-10}m (B) 10^{-20}m (C) 10^{-30}m (D) 10^{-40}m

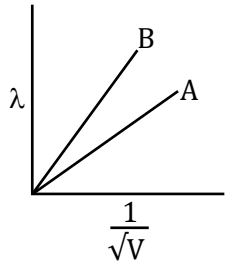
CAS143

63. How fast is an electron moving if it has a wavelength equal to distance travelled in one second –

(A) $\sqrt{\frac{m}{h}}$ (B) $\sqrt{\frac{h}{m}}$ (C) $\sqrt{\frac{h}{p}}$ (D) $\sqrt{\frac{h}{2\times K.E.}}$

CAS222

64. De–Broglie wavelength of two particles A & B are plotted against $\left(\frac{1}{\sqrt{V}}\right)$; where V is potential on the particles. Which of the following relation is correct about mass of particle (charge is same)



(A) $M_A = M_B$ (B) $M_A > M_B$ (C) $M_A < M_B$ (D) $M_A \leq M_B$

CAS223

Heisenberg's Uncertainty Principle

65. The uncertainties in the velocities of two particles, A and B are 0.05 ms^{-1} and 0.02 ms^{-1}, respectively, the mass of B is five times that of the mass of A. What is the ratio of uncertainties in their positions $\left(\frac{\Delta x_A}{\Delta x_B}\right)$

(A) 2 (B) 0.25 (C) 4 (D) 1

CAS145

66. If the electron is to be located within 5×10^{-5} Å, what will be the uncertainty in its velocity ?

(A) 1.16×10^{-10} m/sec (B) 2.16×10^{-10} m/sec

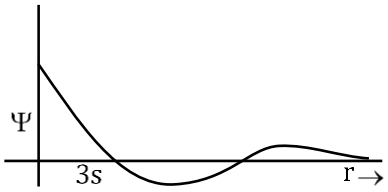
(C) 1.16×10^{10} m/sec (D) 8.16×10^{10} m/sec

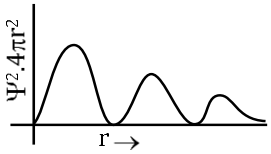
CAS147

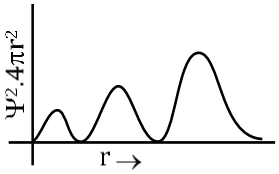
For More Join: @IITJEE_Advanced

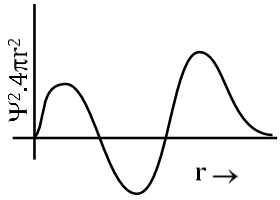
Quantum Mechanical Model of Atom (Schrodinger wave equation using Polar coordinates; Graph transformation; Graphs of Schrodinger wave equation; Equations of different orbitals)

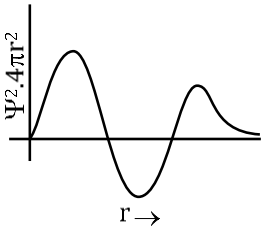
67. The wave function (Ψ) verses radial distance (r) curve for certain orbital is given. Predict the shape of $\Psi^2.4\pi r^2$ (radial probability distribution function) verses r graph.



(A) 

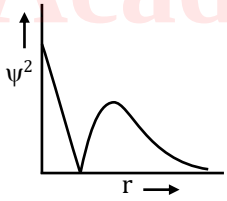
(B) 

(C) 

(D) 

CAS150

68. The following graph between Ψ^2 probability density and distance from the nucleus represents –



(A) 2s (B) 3s (C) 1s (D) 2p

CAS151

69. The radial function for an orbitals is :

$$R(r)=\frac{1}{9\sqrt{6}}\left(\frac{1}{a_0}\right)^{5/2}.(4-\sigma)\,\sigma e^{-\sigma/2};\sigma=\frac{2r}{na_0}.$$

Pick out the incorrect statement :

(A) orbital must be 3p (B) orbital must be $3p_x$

(C) the orbital has one radial node (D) the orbital must have one angular node

CAS152

70. According to Schrodinger model nature of electron in an atom is as :–

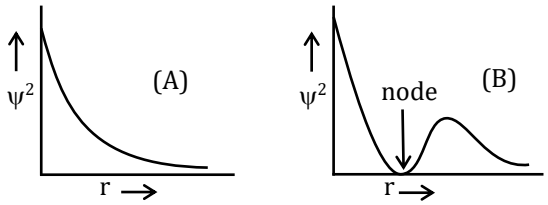
(A) Particles only (B) Wave only

(C) Both simultaneously (D) Sometimes waves and sometimes particle

CAS154

For More Join: @IITJEE_Advanced

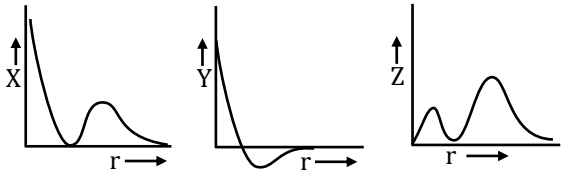
71. In following two plots, ψ^2 is plotted against the distance 'r' from nucleus. Select the correct statement:



(A) 'A' is for 1s and 'B' for 2s
(B) 'A' is for 2s and 'B' for 1s
(C) 'A' is for 2s and 'B' for 2p
(D) 'A' is for 2p and 'B' for 2s

CAS159

72. Plots for 2s orbital are

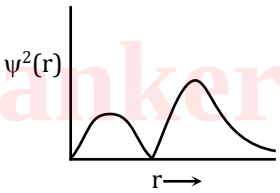


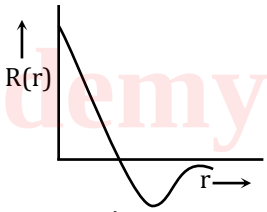
X, Y and Z are respectively
(A) R, R^2 and $4\pi r^2 R^2$
(B) R^2, R and $4\pi r^2 R^2$
(C) $4\pi r^2 R^2$, R^2 and R
(D) R^2, $4\pi r^2 R^2$ and R

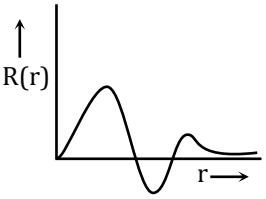
CAS161

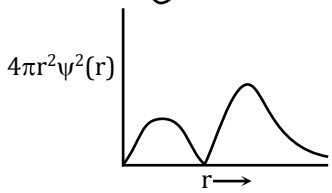
73. The correct curve for 3p orbital is –

Rankers Academy JEE

(A) 

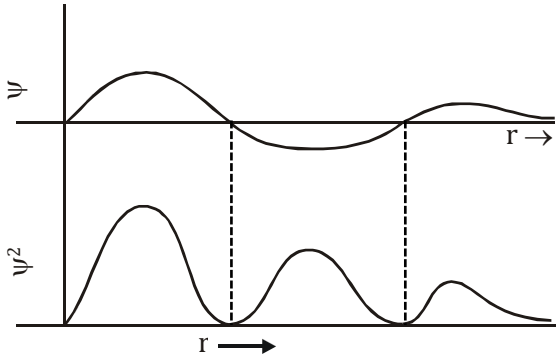
(B) 

(C) 

(D) 

CAS162

74. The wave function ψ vs radial distance (r) and square of wave function $(\psi)^2$ vs radial distance (r) graph of certain orbital is given. Identify the orbital.



(A) 3s (B) 4d (C) 3p (D) 4p

CAS164

For More Join: @IITJEE_Advanced

75. The distance of spherical nodes from nucleus for the given orbital are

$\psi_{radial} = \frac{1}{9\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2} [\sigma^2 - 4\sigma + 3] \exp(-\sigma/2)$

where a_0 & Z are the constants and $\sigma = \frac{2Zr}{na_0}$

(A) Zero, infinity (B) $\frac{a_0}{Z}, \frac{a_0}{2Z}$ (C) $\frac{3a_0}{2Z}, \frac{9a_0}{2Z}$ (D) $\frac{a_0}{Z}, \frac{3a_0}{2Z}$

CAS227

Atomic Orbitals, Shapes of atomic orbitals and nodes/nodal surfaces

76. The orbital angular momentum of an electron in 2s orbital is:

(A) $+\frac{1}{2}.\frac{\hbar}{2\pi}$ (B) Zero (C) $\frac{\hbar}{2\pi}$ (D) $\sqrt{2}.\frac{\hbar}{2\pi}$

CAS224

77. In an excited state, a calcium atom has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 4d^1$. What is the orbital angular momentum for d electron :- $\left(\hbar = \frac{h}{2\pi}\right)$

(A) $\sqrt{4}\hbar$ (B) $\sqrt{16}\hbar$ (C) $\sqrt{6}\hbar$ (D) $\sqrt{10}\hbar$

CAS225

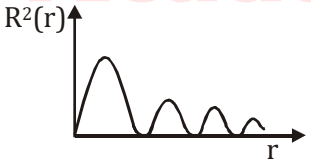
78. From the following observations predict the type of orbital :

Observation 1 : x y plane acts as nodal plane

Observation 2 : The angular function of the orbital intersect the three axis at origin only.

Observation 3 : $R^2(r)$v/s r curve is obtained for the orbital is

Rankers Academy JEE



(A) $5p_z$ (B) $6d_{xy}$ (C) $6d_{x^2-y^2}$ (D) $6\ d_{yz}$

CAS163

Quantum numbers

79. What would be the approximate quantum number, n, for a circular orbit of hydrogen, 1×10^{-5} cm in diameter ?

(A) 31 (B) 43 (C) 40 (D) 39

CAS128

80. A photon of energy 12.75 ev is completely absorbed by a hydrogen atom initially in ground state. The principle quantum number of the excited state is

(A) 1 (B) 3 (C) 4 (D) ∞

CAS204

81. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number + 2 :

(A) 3 (B) 4 (C) 2 (D) 1

CAS221

For More Join: @IITJEE_Advanced

82. Which of the following sets of quantum numbers represent an impossible arrangement :

(A) n = 3, l = 2, m = –2, $m_s = \frac{1}{2}$ (B) n = 4, l = 0, m = 0, $m_s = \frac{1}{2}$

(C) n = 3, l = 2, m = –3, $m_s = \frac{1}{2}$ (D) n = 5, l = 3, m = 0, $m_s = \frac{1}{2}$

CAS012

Rules for Electronic Configuration; Aufbau's Principle; Pauli's Exclusion Principle; Hund's Rule; Half and fully filled subshell configuration; All exceptional configuration; Degeneracy

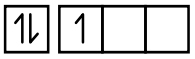
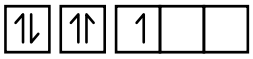
83. The decreasing order of energy of the 3d, 4s, 3p, 3s orbitals is :

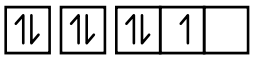
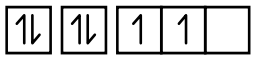
(A) 3d > 3s > 4s > 3p (B) 3s > 4s > 3p > 3d

(C) 3d > 4s > 3p > 3s (D) 4s > 3d > 3s > 3p

CAS014

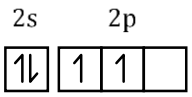
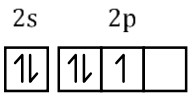
84. Which configuration does not obey Pauli's exclusion principle :-

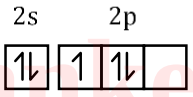
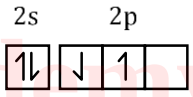
(A)  (B) 

(C)  (D) 

CAS169

85. Which of the following configuration follows the Hund's rule :–

(A)  (B) 

(C)  (D) 

CAS170

86. The basis of three unpaired electrons present in the configuration of nitrogen is :–

(A) Aufbau principle (B) Pauli's principle

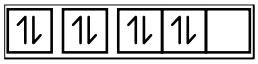
(C) Hund's principle (D) Uncertainty principle

CAS171

87. No. of all subshells having n + ℓ = 7 is:-

(A) 4 (B) 5 (C) 6 (D) 7

CAS173

88. Electronic configuration  has violated :–

(A) Hund's rule (B) Pauli's principle (C) Aufbau principle (D) (n + ℓ) rule

CAS174

89. The total spin resulting from a d^9 configuration is:-

(A) $\frac{1}{2}$ (B) 2 (C) 1 (D) $\frac{3}{2}$

CAS175

90. A neutral atom of an element has two K, eight L, nine M and two N electrons then electronic configuration of the element is

(A) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^1$ (B) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$

(C) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3d^2\ 3p^6\ 4s^1$ (D) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$

CAS176

Rankers Academy JEE

For More Join: @IITJEE_Advanced

91. In ground state of $_{24}Cr$, number of unpaired electron :-

(A) 10 (B) 12

(C) 6 (D) 18

CAS177

92. In Palladium (Atomic no.–46), number of electron having ($\ell = 2$) will be -

(A) 20 (B) 18

(C) 16 (D) 22

CAS341

93. The number of electrons in Ca having minimum value of $\left|\frac{n}{\ell \times m_\ell}\right|$ is.

(Consider only non-zero values of ℓ and m)

(A) 6 (B) 3

(C) 4 (D) None of these

CAS342

94. Which of the following pair is correct for isosters :-

(A) N_2O, CH_4 (B) N_2O, CO_2

(C) CO_2, SO_2 (D) N_2O, HOCl

CAS343

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Bohr's Model for Hydrogen Atom

1. Choose the incorrect statement(s):

(A) Increasing order of wavelength is
Micro waves > Radio waves > IR waves > visible waves > UV waves

(B) The order of Bohr radius is (r_n : where n is orbit number for a given atom) $r_1 < r_2 < r_3 < r_4$

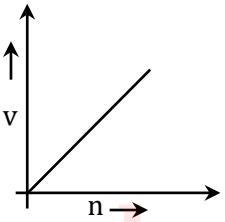
(C) The order of total energy is (E_n : where n is orbit number for a given atom) $E_1 > E_2 > E_3 > E_4$

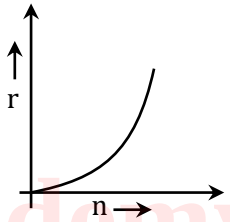
(D) The order of velocity of electron in H, He^+, Li^+, Be^{3+} species in second Bohr orbit is
$Be^{3+} > Li^{+2} > He^+ > H$

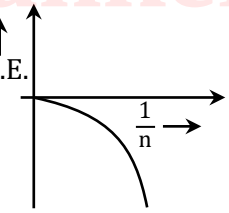
CAS230

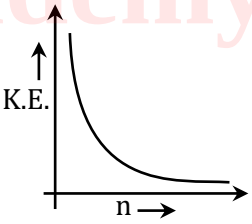
2. Select the correct curve(s):

If v = velocity of electron in Bohr's orbit
r = Radius of electron in Bohr's orbit
P.E. = Potential energy of electron in Bohr's orbit
K.E. = Kinetic energy of electron in Bohr's orbit.

(A) 

(B) 

(C) 

(D) 

Rankers Academy JEE

CAS231

3. Which is / are correct statement.

(A) The difference in angular momentum associated with the electron present in consecutive orbits of H–atom is $(n-1)\frac{h}{2\pi}$.

(B) Energy difference between energy levels will be changed if, P.E. at infinity assigned value other than zero.

(C) Frequency of spectral line in a H–atom is in the order of $(2 \to 1) < (3 \to 1) < (4 \to 1)$.

(D) On moving away from the nucleus, kinetic energy of electron decreases.

CAS232

Explanation of Line Spectrum of Hydrogen

4. In a H–like sample electrons make transition from 5th excited state to 2nd excited state

(A) 10 different spectral lines will be emitted

(B) 6 different spectral lines will be emitted

(C) Number of lines belonging to Balmer series will be 4

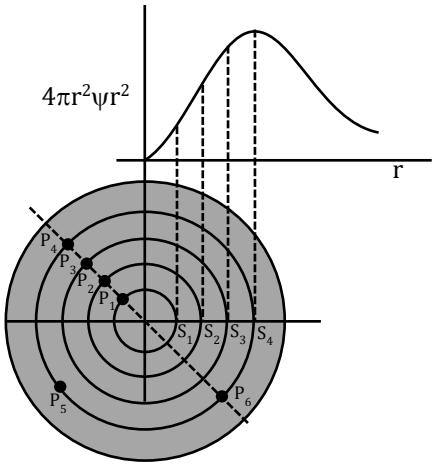
(D) Number of lines belonging to Paschen series will be 3

CAS229

For More Join: @IITJEE_Advanced

Quantum Mechanical Model of Atom (Schrodinger wave equation using Polar coordinates; Graph transformation; Graphs of Schrodinger wave equation; Equations of different orbitals)

5. For s–orbital, $4\pi r^2 \psi^2_r$ vs r is plotted. Select correct statements (s)



Note : S = spherical surface at given distance

P = point (very small volume) at given distance

(A) Order of probability of finding an electron at points is $P_4 > P_3 > P_2 > P_1$

(B) Order of probability of finding an electron at spherical surface is $S_4 > S_3 > S_2 > S_1$

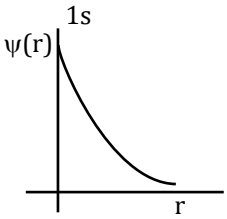
(C) Order of probability of finding an electron at points is $P_6 = P_5 = P_4$

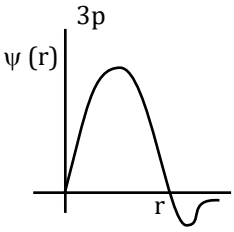
(D) Order of probability of finding an electron at points is $P_4 = P_3 = P_2 = P_1$

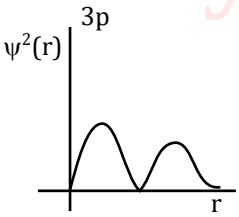
CAS155

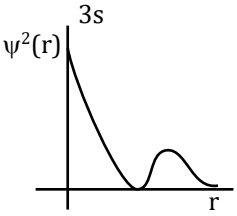
Rankers Academy JEE

6. Which of the following radial distribution curves is correctly plotted.

(A) 

(B) 

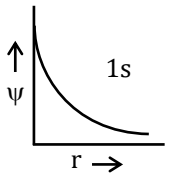
(C) 

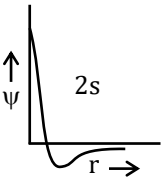
(D) 

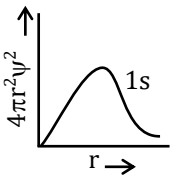
CAS157

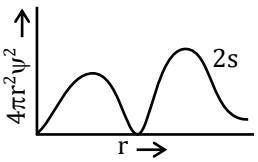
7. Which of the following graphs are correct ?

NOTE : r is distance from the nucleus

(A) 

(B) 

(C) 

(D) 

CAS158

Rules for Electronic Configuration; Aufbau's Principle; Pauli's Exclusion Principle; Hund's Rule; Half and fully filled subshell configuration; All exceptional configuration; Degeneracy

8. For an electron present in which of the following orbital for which (n + *l*) value is maximum.

(A) 3p (B) 5p (C) 4d (D) 5s

CAS344

For More Join: @IITJEE_Advanced

9. The quantum numbers for the 19th electron of Cr (Z = 24) are

(A) n = 3, ℓ = 0, m = 0, m_s = + ½ (B) n = 4, ℓ = 0, m = 0, m_s = + ½

(C) n = 3, ℓ = 2, m = 2, m_s = + ½ (D) n = 4, ℓ = 0, m = 0, m_s = – ½

CAS346

10. The species which have same number of electrons in outer most and penultimate shell -

(A) Ca (B) Ar (C) V^{+3} (D) Sc^{3+}

CAS347

11. Which sub-shell fill completely before the 4f ?

(A) 6s (B) 5p (C) 5d (D) 4d

CAS348

12. Select incorrect statement(s) :

(A) orbital has different shape from rest of all d-orbitals

(B) For the formation of cation electrons are always removed from 4s.

(C) Zinc is a p-block element.

(D) Principal quantum number depend upon the value of azimuthal quantum number

CAS349

MATRIX MATCH TYPE QUESTION

Towards Quantum Mechanical Model of Atom; De Broglie's Hypothesis

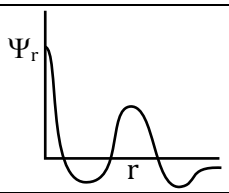
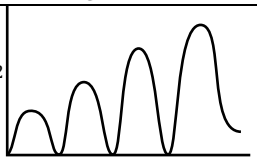
13.

Column–I	Column–II
(A) Electron moving in 2nd orbit in He^+ ion	(P) Radius of orbit in which electron is moving is 0.529 Å
(B) Electron moving in 3rd orbit in H–atom	(Q) Total energy of electron is (–)13.6 × 9eV
(C) Electron moving in 1st orbit in Li^{+2} ion	(R) Velocity of electron is $\frac{2.188\times10^6}{3}$ m/sec
(D) Electron moving in 2nd orbit is Be^{+3} ion	(S) De–Broglie wavelength of electron is $\sqrt{\frac{150}{13.6}}$ Å

CAS233

Quantum Mechanical Model of Atom (Schrodinger wave equation using Polar coordinates; Graph transformation; Graphs of Schrodinger wave equation; Equations of different orbitals)

14. Column I & Column II contain data on Schrodinger Wave–Mechanical model, where symbols have their usual meanings. Match the columns.

Column I	Column II (Type of orbital)
(A) 	(P) 4s
(B) $\Psi_r^2 4\pi r^2$ 	(Q) $5p_x$
(C) $\Psi(\theta,\phi)$ = K (independent of θ & ϕ)	(R) 3s
(D) at least one angular node is present	(S) $6d_{xy}$

CAS234

For More Join: @IITJEE_Advanced

Quantum numbers

15. **Column-I** **Column-II**

Element **Max. value of n and ℓ respectively ; consider filled subshell only**

(P) P (1) 6, 2
(Q) N (2) 3, 1
(R) Pb (3) 6, 3
(S) Cs (4) 2, 1

Code :

	P	Q	R	S
(A)	4	1	2	3
(B)	2	4	1	3
(C)	3	1	2	4
(D)	2	4	3	1

CAS332

Rules for Electronic Configuration; Aufbau's Principle; Pauli's Exclusion Principle; Hund's Rule; Half and fully filled subshell configuration; All exceptional configuration; Degenerac

16. Match the following:

Column–I	**Column–II**
(P) Same number of unpaired electrons are present in (excluding zero)	(1) Na^+, Mg^{2+}, F^-
(Q) Same number of electrons in s & p subshells.	(2) F^-, Mg, O^{2-}
(R) Same number of electrons with the $l = 1$	(3) Mg, Ne, O^{2-}
(S) Same number of total electrons	(4) Li, Na, K

Code :

	P	Q	R	S
(A)	1, 2	3, 4	1	4
(B)	4, 2	3, 1	2, 3	4, 1
(C)	4	1	1, 2, 3	1
(D)	3	1, 2	3, 4	1, 3, 4

CAS350

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

Column- I	Column - II	Column - III
(A) Paramagnetic set	(i) Na^+, Mg^{+2}, F^-	(P) same value of principal quantum number for last electron
(B) Isoelectronic set	(ii) Li, Na, K	(Q) The non zero number(s) of e^- (s) for n = 3 and $\ell \geq 1$ is
(C) The set for which value(s) of spin multiplicity is ≥ 1	(iii) Fe^{+3}, Co^{2+}, Ni^{+2}	(R) The value of "m_s" must be + $\frac{1}{2}$ for last electron
(D) The set of elements belongs to same period in periodic table	(iv) S^{-2}, Cl^-, P^{-3}	(S) Set for which the value of m = ±2 is possible for electron(s)

Rankers Academy JEE

For More Join: @IITJEE_Advanced

17. Which one of the following options is the **CORRECT** combination?
(A) (A, i, P) (B) (B, iv, S) (C) (D, iii, S) (D) (C, iii, R)

CAS351

18. Which one of the following options is the **INCORRECT** combination?
(A) (A, iii, P) (B) (C, ii, P) (C) (B, i, P) (D) (B, iv, Q)

CAS352

19. Which one of the following options is the **CORRECT** set of species with number of nodal planes for filled/partially filled orbitals is 1 for all given species in set?
(A) (B, ii, R) (B) (B, iv, P) (C) (A, i, Q) (D) (D, iii, S)

CAS353

COMPREHENSION TYPE QUESTIONS

Bohr's Model for Hydrogen Atom

Paragraph for Q. No. 20 to 23 (4 questions)

The only electron in the hydrogen atom resides under ordinary conditions on the first orbit. When energy is supplied, the electron moves to higher energy orbit depending on the amount of energy absorbed. When this electron returns to any of the lower orbits, it emits energy. Lyman series is formed when the electron returns to the lowest orbit while Balmer series is formed when the electron returns to second orbit. Similarly, Paschen, Brackett and Pfund series are formed when electron returns to the third, fourth and fifth orbits from higher energy orbits respectively.

Maximum number of lines produced when an electron jumps from n^{th} level to ground level is equal to $\frac{n(n-1)}{2}$. For example, in the case of n = 4, number of lines produced is 6. ($4 \to 3$, $4 \to 2$, $4 \to 1$, $3 \to 2$, $3 \to 1$, $2 \to 1$). When an electron returns from n_2 to n_1 state, the number of lines in the spectrum will be equal to

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

If the electron comes back from energy level having energy E_2 to energy level having energy E_1, then the difference may be expressed in terms of energy of photon as:

$$E_2 - E_1 = \Delta E, \lambda = \frac{hc}{\Delta E}$$

Since h and c are constants, ΔE corresponds to definite energy; thus, each transition from one energy level to another will produce a light of definite wavelength. This is actually observed as a line in the spectrum of hydrogen atom.

Wave number of lines is given by the formula $\bar{\nu} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$.

where R is a Rydberg's constant ($R = 1.1 \times 10^7\ m^{-1}$)

20. The energy of photon emitted corresponding to transition n = 3 to n = 1 is [$h = 6 \times 10^{-34}$ J–sec.]
(A) 1.76×10^{-18} J (B) 1.98×10^{-18} J (C) 1.76×10^{-17} J (D) None of these

CAS235

Rankers Academy JEE
For More Join: @IITJEE_Advanced

21. In a collection of H–atom, electrons make transition from 5th excited state to 2nd excited state then maximum number of different types of photons observed are

(A) 3 (B) 4 (C) 6 (D) 15

CAS236

22. The difference in the wavelength of the 1st line of Lyman series and 2nd line of Balmer series in a hydrogen atom is

(A) $\frac{9}{2R}$ (B) $\frac{4}{R}$ (C) $\frac{88}{15R}$ (D) None

CAS237

23. The wave number of electromagnetic radiations emitted during the transition of electron in between two levels of Li^{2+} ion whose principal quantum numbers sum is 4 and difference is 2 is

(A) 3.5 R (B) 4 R (C) 8 R (D) $\frac{8}{9}R$

CAS238

Heisenberg's Uncertainty Principle

Paragraph for Q. No. 24 to 26 (3 questions)

The French physicist Louis de Broglie in 1924 postulated that matter, like radiation, should exhibit a dual behaviour. He proposed the following relationship between the wavelength λ of a material particle, its linear momentum p and Planck constant h.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Rankers Academy JEE

The de Broglie relation implies that the wavelength of a particle should decreases as its velocity increases. It also implies that for a given velocity heavier particles should have shorter wavelength than lighter particles. The waves associated with particles in motion are called matter waves or de Broglie waves. These waves differ from the electromagnetic waves as they

(i) have lower velocities

(ii) have no electrical and magnetic fields and

(iii)are not emitted by the particle under consideration.

The experimental confirmation of the de Broglie relation was obtained when Davisson and Germer, in 1927, observed that a beam of electrons is diffracted by a nickel crystal. As diffraction is a characteristic property of waves, hence the beam of electron behaves as a wave, as proposed by de Broglie.

Werner Heisenberg considered the limits of how precisely we can measure properties of an electron or other microscopic particle like electron. He determined that there is a fundamental limit of how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the "Heisenberg uncertainty principle: It is impossible to determine simultaneously and precisely both the momentum and position of a particle. The product of uncertainty in the position, Δx and the uncertainty in the momentum $\Delta(mv)$ must be greater than or equal to $\frac{h}{4\pi}$. i.e.

$$\Delta x\, \Delta(mv) \geq \frac{h}{4\pi}$$

For More Join: @IITJEE_Advanced

24. The correct order of wavelength of Hydrogen ($_1H^1$), Deuterium ($_1H^2$) and Tritium ($_1H^3$) moving with same kinetic energy is

(A) $\lambda_H > \lambda_D > \lambda_T$ (B) $\lambda_H = \lambda_D = \lambda_T$

(C) $\lambda_H < \lambda_D < \lambda_T$ (D) $\lambda_H < \lambda_D > \lambda_T$

CAS239

25. The transition, so that the de–Broglie wavelength of electron becomes 3 times of its initial value in $He^{\oplus}$ ion will be

(A) $2 \rightarrow 5$ (B) $3 \rightarrow 2$

(C) $2 \rightarrow 6$ (D) $1 \rightarrow 2$

CAS240

26. If the uncertainty in velocity & position is same, then the uncertainty in momentum will be

(A) $\sqrt{\frac{hm}{4\pi}}$ (B) $m\sqrt{\frac{h}{4\pi}}$

(C) $\sqrt{\frac{h}{4\pi m}}$ (D) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

CAS241

Atomic Orbitals, Shapes of atomic orbitals and nodes/nodal surfaces

Paragraph for Question 27 & 28

Consider a hypothetical atom where p_x, p_y, p_z, d_{xy}, d_{xz}, d_{yz} and $d_{x^2-y^2}$ orbitals are present for principal quantum number n = 4.

27. Find the number of other orbital which lobes are fully present in the nodal plane of p_x orbital :-

(A) 2 (B) 4 (C) 3 (D) 5

CAS354

28. Which of the following orbital's lobe is not present at all, either in the nodal plane of p_x orbital or in the nodal plane of p_y orbital.

(A) d_{xy} (B) d_{yz} (C) $d_{x^2-y^2}$ (D) p_z

CAS355

Bankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Discovery of subatomic particles (Discharge Tube Experiment; Cathode rays) charge to mass ratio(e/m)

1. Specific charges of two particles A and B are in ratio 2 : 3. If their mass ratio $m_A : m_B$ is 2 : 3, then find the value of $\frac{18}{8}\times\left(\frac{q_A}{q_B}\right)$?

CAS102

Anode Rays; Discovery of proton, neutron

2. The ratio of specific charges $\left(\frac{q}{m}\right)$ of an α–particle and a deuteron is x : 1.00. The value of 'x' is.

CAS242

Atomic models (Thomson Model of Atom , Rutherford Nuclear Model of Atom)

3. If the radius of an atom is 2Å and the radius of its nucleus is 4×10^{-15} m. The ratio of volume of nucleus to that of atom is x : 10^{15}. The value of 'x' is.

CAS244

Developments leading to the Bohr's model of atom; Maxwell's Electromagnetic Wave Theory; Wave characteristics such as wavelength, frequency, time period, wave number; Electromagnetic Spectrum

Pathfinders Academy JEE

4. The wave numbers of two electromagnetic radiations are 4×10^6 m^{-1} and 2×10^5 cm^{-1}. The ratio of their frequencies is $\frac{p}{q}$. Then the value of $20 \times \frac{p}{q}$ is

CAS113

5. A radio station emits the radiations of wavelength 2000 m (meter band of station). What is the frequency band of that station, in kHz (Frequency of emitted radio waves) ?

CAS248

Particle Nature of Electromagnetic Radiation: Planck's Quantum Theory

6. Find the number of photons (in the order of 10^{22}) of radiation of frequency 5×10^{13} s^{-1} that must be absorbed in order to melt one gm ice when the latent heat of fusion of ice is 330 J/g.
(h = 6.6×10^{-34} J × sec.)

CAS250

7. The quantum yield for decomposition of HI is 0.2. In an experiment 0.01 moles of HI are decomposed. Find the number of photons (in the order of 10^{22}) absorbed. ($N_A = 6 \times 10^{23}$)

CAS252

8. The dissociation energy of H_2 is 482.5 KJ/mol. If H_2 is exposed to radiant energy of wavelength 124 nm, what % of radiant energy will be converted into K.E.

($1\frac{ev}{atom}$ = 96.5 kJ/mol, hc = 1240 eV × nm)

CAS253

For More Join: @IITJEE_Advanced

Photoelectric Effect

9. The minimum energy required for the emission of a metal electron is 13.24×10^{-19} J. Then the critical frequency of the photon (threshold frequency) required to eject the electron is $y \times 10^{15}$. Then the value of y is

CAS121

10. A light source of wavelength λ illuminates a metal and ejects photo electron with $(KE)_{max} = 1$ eV. Another light source of wave length $\frac{\lambda}{3}$, ejects photoelectrons from same metal with $(KE)_{max} = 5$eV. Find the value of work function (eV) of metal.

CAS255

Bohr's Model for Hydrogen Atom

11. If the ratio of radius of 2nd excited state of H & 1st excited state of $Li^{2\oplus}$ is $\frac{p}{q}$. Then the value of $\frac{4}{3} \times \frac{p}{q}$ is

CAS134

12. The ionisation energy for a single electron system is 14.4 eV. Calculate the amount of energy (in eV) released when electron jumps from 3rd orbit to 2nd orbit.

CAS135

13. If an electron in H–atom jumps from one orbit to other its angular momentum doubles. If the new radius is 'x' times the old radius, then find x ?

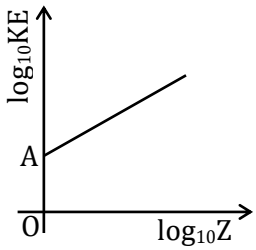
CAS259

14. Calculate the frequency of revolution of electron (in 10^{14} revolutions per sec) in the first Bohr orbit in a H–atom.

Rankers Academy JEE

CAS263

15. In a H–like atom for an electron revolving in n^{th} orbit, the variation of $\log_{10}$ **KE** versus $\log_{10}$**Z** is plotted as following –



where Z is atomic number & KE is kinetic energy of electron (in eV). If OA = $\log_{10}$ 3.4 then find the value of principle quantum number of the electron.

CAS266

16. The energy of the second stationary state in Li^{+2} ion is -4.95×10^{-18} J. Find ionization energy (in KJ/mole) for He^{+} ions in its ground state ($N_A = 6 \times 10^{23}$).

CAS267

17. A hydrogen like atom with atomic number 'Z' is in higher excited state of quantum number 'n'. This excited state atom can make a transition to the first excited state by successively emitting two photons of energies 10.2 eV and 17.0 eV respectively. Alternatively, the atom from the same excited state can make a transition to the 2nd excited state by emitting two photons of energies 4.25 eV and 5.95 eV respectively. Calculate the value of 'Z'.

CAS269

For More Join: @IITJEE_Advanced

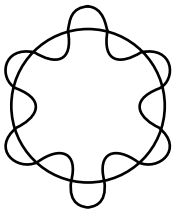
Explanation of Line Spectrum of Hydrogen

18. In H–spectrum, longest wavelength of Lyman is 120 nm and shortest wavelength of Balmer is 360 nm. From this data, find longest wavelength (in nm) of photon, that can ionize this H–atom

CAS274

Towards Quantum Mechanical Model of Atom; De Broglie's Hypothesis

19. The figure shows a sample of H–atoms having electron revolving in higher orbit 'n'.



If this electron makes transition from this orbit 'n' to ground state, number of Paschen lines emitted are.

CAS279

20. de–Broglie wavelength associated with an electron in 4th orbit of hydrogen atom is a × (πr_0) where r_0 is radius of 1st orbit of hydrogen atom, find value of 'a'.

CAS280

Heisenberg's Uncertainty Principle

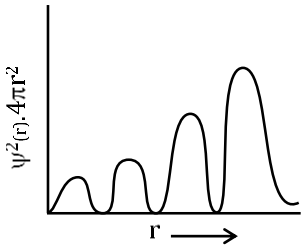
21. If uncertainties in measurement of position and momentum of an electron are equal, then the uncertainty in measurement of its speed is x × 10^{12}. The value of x (the closest nearest integer value) is $m_e = 9.1 \times 10^{-31}$ kg, and h = 6.625×10^{-34} Js.

Rankers Academy JEE

CAS149

Quantum Mechanical Model of Atom (Schrodinger wave equation using Polar coordinates; Graph transformation; Graphs of Schrodinger wave equation; Equations of different orbitals)

22. An atomic orbital of H atom has the following characteristics. Calculate sum of radial & angular nodes. Characteristics I → The graph of radial probability distribution looks like.



Characteristics II → The shape of the orbital is double dumb–bell.

CAS165

23. For a certain orbital

$$\Psi(r)=\frac{1}{16\sqrt{4}\,a_0^{\frac{3}{2}}}(1-\sigma)\left(\sigma^2-8\sigma+12\right)e^{-\frac{\sigma}{2}},\sigma=\frac{2r}{na_0}\ \{a_0 = \text{constant}\}$$

The distance of nearest radial node from nucleus is xa_0, while farthest radial node from nucleus is ya_0. Find $\left(\frac{y}{x}\right)$.

CAS166

For More Join: @IITJEE_Advanced

24. The ratio of angular nodes of 3d orbital and radial nodes of 3p orbital is x : 1 what is the value of x.

CAS168

25. How many non–spherical subshell are possible that have at least two maxima if a curve is plotted between radial probability distribution function versus radial distance for which principal quantum number : $n \leq 4$?

CAS284

Rules for Electronic Configuration; Aufbau's Principle; Pauli's Exclusion Principle; Hund's Rule; Half and fully filled subshell configuration; All exceptional configuration; Degeneracy

26. Minimum number of electrons having $m_s = \left(-\frac{1}{2}\right)$ in Cr is "____".

CAS356

27. How many elements are possible for the I^{st} period of periodic table if azimuthal quantum number can have integral values from 0 to (n + 1). [n = shell number & other rules are remaning same to form periodic table.

CAS357

28. Find number of unpaired electrons when Fe does not follow $(n + \ell)$ rule and filling of electron takes place shell after shell and Hund's rule is also not obeyed.

CAS358

29. Find the sum of maximum unpaired e^- present in one 5g & one 6g orbital.

CAS360

30. Find the sum of minimum and maximum possible value of x in Fe^{+x} ion, if magnetic moment of Fe^{+x} = 4.89 B.M.

CAS361

Bankers Academy JEE

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Atomic models (Thomson Model of Atom , Rutherford Nuclear Model of Atom)

31. The kinetic energy of a charged particle (charge = q) accelerated by a potential difference of 10000 volt is 1.6×10^{-15} Joule. The value of 'q' (in 10^{-19} coulomb) is.

CAS243

32. An α–particle of K.E. 5.4 MeV is projected towards Cr–nucleus (Z = 24). What is its distance of closest approach (in 10^{-14} m) ? ($e = 1.6 \times 10^{-19}$C)

CAS103

33. The ratio of distances of closest approach of a proton and an α–particle projected towards the same nucleus with the same initial kinetic energy is x : 1. The value of 'x' is.

CAS245

For More Join: @IITJEE_Advanced

34. With what speed (in 10^7 ms^{-1}) an α–particle should be projected towards stationary krypton nucleus (Z = 36) such that it approaches upto 2.4×10^{-14} m from the nucleus ?
($N_A = 6 \times 10^{23}$, $e = 1.6 \times 10^{-19}C$).

CAS246

Developments leading to the Bohr's model of atom; Maxwell's Electromagnetic Wave Theory; Wave characteristics such as wavelength, frequency, time period, wave number; Electromagnetic Spectrum

35. The wave numbers of two electromagnetic radiations are 4×10^6 m^{-1} and 2×10^5 cm^{-1}. The ratio of their frequencies is x : 1.00. the value of 'x' is.

CAS021

36. A potential difference of 20 kV is applied across an X-ray tube. Find the minimum wavelength (in Å) of X-ray generated.

CAS025

37. An electromagnetic radiation makes 5000 waves in 20 cm. The frequency of radiation (in 10^{12} Hz) is.

CAS026

38. How long (sec.) would it take a radio wave of frequency 6×10^3 sec^{-1} to travel from mars to the earth, a distance of 8×10^5 km ?

CAS027

Particle Nature of Electromagnetic Radiation: Planck's Quantum Theory

39. A certain dye absorbs 4000 Å and fluoresces at 5000 Å these being wavelengths of maximum absorption that under given conditions 40% of the absorbed energy is emitted. If the ratio of the number of quanta emitted to the number absorbed is x : 1.00, the value of 'x' is.

Rankers Academy JEE

CAS251

Photoelectric Effect

40. The K.E. of an electron emitted from tungsten surface is 3.06 eV. What voltage would be required to bring the electron to rest.

CAS120

41. The K.E. of an electron emitted from tungsten surface is 3.06 eV. What voltage (in volt) would be required to bring the electron to rest.

CAS256

42. Photon having wavelength 12.4 nm was allowed to strike a metal plate having work function 25 eV. Calculate the
(a) Maximum kinetic energy (in eV) of photoelectrons emitted in eV.
(b) Wavelength (in Å) of electron with maximum kinetic energy in Å.
[Given : hc = 1240 nm.eV]

CAS257

Bohr's Model for Hydrogen Atom

43. If the mass of electron is doubled, then find the new radius (in Å) of 1st orbit of H–atom ?

CAS258

44. The radius of the an orbit of hydrogen atom is 0.85 nm. Calculate the speed (in 10^5 m/sec) of electron in this orbit.

CAS260

For More Join: @IITJEE_Advanced

45. If the average life time of an excited state of H atom is of order 10^{-8} sec, estimate how many revolutions (in the order of 10^6) an e^- makes when it is in the state n = 2 and before it suffers a transition to n =1 state.

CAS262

46. Calculate energy (in eV) of electron which is moving in the orbit of a hydrogen atom that has its radius, sixteen times the radius of first Bohr orbit for H–atom.

CAS264

47. H–atom is exposed to electromagnetic radiation of 1026 Å and gives out induced radiations (radiations emitted when e^- returns to ground state). Calculate λ (in Å) of induced radiations.

CAS270

48. Suppose the potential energy between electron and proton at a distance r is given by $-\frac{ke^2}{3r^3}$. Use Bohr's theory to obtain energy of such a hypothetical atom.

CAS271

49. In a hypothetical H–atom, the mass of electron & its charge is double of what we consider, then calculate the total energy (in eV) of electron in the I^{st} orbit of such a hypothetical H–atom ?
[Assuming all others concepts and parameter to be same as we considered in Bohr's model.]

CAS273

50. The angular momentum of an electron in a Bohr's orbit of H–atom is 3.1652×10^{-34} kg–m^2/sec. Calculate the wavenumber (in 10^6 m^{-1}) of the spectral line emitted when an electron falls from this level to the ground state. **[h = 6.626 × 10^{-34} Js, R = 1.09 × 10^7 m^{-1}]**

CAS276

Pankers Academy JEE

51. The ionisation energy of the hydrogen atom is given to be 13.6 eV. A photon falls on a hydrogen atom which is initially in the ground state and excites it to the (n = 4) state. Calculate the wavelength (in Å) of the photon (hc = 12400 Å.eV).

CAS277

52. The ionisation energy of a H–like Bohr atom is 4 Rydberg's.
(i) What is the wavelength (in Å) of radiation emitted when the electron jumps from the first excited state to the ground state.
(ii) What is the radius (in Å) of first Bohr orbit for this atom.
[1 Rydberg = 2.20×10^{-18} J, h = 6.60×10^{-34} J.s]

CAS278

Towards Quantum Mechanical Model of Atom; De Broglie's Hypothesis

53. If $y \times 10^{-4}$ V potential difference must an electron pass to have a wavelength of 500 Å. Then find value of y.

CAS029

Heisenberg's Uncertainty Principle

54. A proton is accelerated to one– tenth of the speed of light. If its velocity can be measured with a precision $\pm$ 1%. What must be its minimum uncertainty in position (in 10^{-12} m).

(Use : $\frac{h}{4\pi}$ = 5.28 × 10^{-35}) (Mass of proton = 1.6 × 10^{-24} gm)

CAS148

For More Join: @IITJEE_Advanced

55. The uncertainty in the position (in cm) of an electron (mass 9.1×10^{-28}gm) moving with a speed of 3×10^4 cm sec^{-1}, uncertainty in speed is 0.011% will be : **($h = 6.6 \times 10^{-34}$ J × s)**

CAS281

56. The diameter of a dust particle of mass 10^{-3} gm is 2Å. If uncertainty in speed of this particle is $\frac{3.31}{\pi} \times 10^{-3}\, m/s$, then find minimum uncertainty in its position (in 10^{-25} m)

($h = 6.62 \times 10^{-34}$ J.s.)

CAS283

Quantum Mechanical Model of Atom (Schrodinger wave equation using Polar coordinates; Graph transformation; Graphs of Schrodinger wave equation; Equations of different orbitals)

57. Calculate the longest distance (in Å) of spherical nodes for '3s' orbital from nucleus in hydrogen atom?

$$R_{3s} = \frac{1}{9\sqrt{3a_0^{3/2}}}(6 - 6\sigma + \sigma^2)\, e^{\frac{-\sigma}{2}} \qquad \text{where } \sigma = \frac{2r}{na_0}$$

CAS167

58. For an orbital in B^{+4} radial function is :

$$R(r) = \frac{1}{9\sqrt{6}}\left(\frac{z}{a_0}\right)^{\frac{3}{2}}(4-\sigma)\sigma\, e^{-\sigma/2}$$

where $\sigma = \frac{2Zr}{na_0}$ and $a_0 = 0.529$ Å ; Z = atomic number, r = radial distance from nucleus.

The radial node of orbital is at distance ..... Å from nucleus.

Rankers Academy JEE

CAS285

Quantum numbers

59. The velocity of e^- in a certain Bohr orbit of the hydrogen atom bears the ratio 1:275 to the speed of light.

(i) What is the quantum no. "n" of the orbit.

(ii) Calculate the wave number (in 10^6 m^{-1}) of the radiation emitted for the transition from the quantum state (n+1) to the ground state.

CAS261

60. A H–like ion is observed to emit six different wavelengths originating from all possible transitions between a group of levels. These levels have energies between –0.85 eV (Min) and –0.544 eV (Max).

(a) Find atomic number (z) of element

(b) Find quantum numbers of levels between which transitions occur.

(c) Calculate the energy (eV) of the longest wavelength emitted in transitions between the levels.

CAS275

For More Join: @IITJEE_Advanced

EXERCISE - (JEE (Main) PYQ)

1. Amongst the following statements, that which was not proposed by Dalton was: **[JEE (Main) 2020]**

(1) all the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.
(2) chemical reactions involve reorganisation of atoms. These are neither created nor destroyed in a chemical reaction.
(3) when gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P.
(4) matter consists of indivisible atoms.

CAS043

2. The energy of one mole of photons of radiation of frequency 2×10^{12} Hz in $J\,mol^{-1}$ is _____.
(Nearest integer)
(Given: $h = 6.626 \times 10^{-34}\,Js$; $N_A = 6.022 \times 10^{23}\,mol^{-1}$) **[JEE (Main) 2023]**

CAS180

3. A 50 watt bulb emits monochromatic red light of wavelength of 795 nm. The number of photons emitted per second by the bulb is $x \times 10^{20}$. The value of x is _____.
[Given : $h = 6.63 \times 10^{-34}$ Js and $c = 3.0 \times 10^{8}$ ms^{-1}] **[JEE (Main) 2021]**

CAS049

4. The minimum energy that must be possessed by photons in order to produce the photoelectric effect with platinum metal is:
[Given: The threshold frequency of platinum is 1.3×10^{15} s^{-1} and $h = 6.6 \times 10^{-34}$ J s.**]** **[JEE (Main) 2022]**

(1) 3.21×10^{-14} J
(2) 6.24×10^{-16} J
(3) 8.58×10^{-19} J
(4) 9.76×10^{-20} J

CAS050

5. Given below are two statements : **[JEE (Main) 2025]**
Statement (I) : A spectral line will be observed for a $2p_x \rightarrow 2p_y$ transition.
Statement (II) : $2p_x$ and $2p_y$ are degenerate orbitals.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) Both **Statement-I** and **Statement-II** are true.
(2) Both **Statement-I** and **Statement-II** are false.
(3) **Statement-I** is true but **Statement-II** is false.
(4) **Statement-I** is false but **Statement-II** is true.

CAS434

6. Number of spectral lines obtained in He^+ spectra, when an electron makes transition from fifth excited state to first excited state will be **[JEE (Main) 2024]**

CAS334

7. Radius of the first excited state of Helium ion is given as :
$a_0 \rightarrow$ radius of first stationary state of hydrogen atom. **[JEE (Main) 2025]**

(1) $r = \frac{a_0}{2}$
(2) $r = \frac{a_0}{4}$
(3) $r = 4a_0$
(4) $r = 2a_0$

CAS435

Parkers Academy JEE
For More Join: @IITJEE_Advanced

8. If a_0 is denoted as the Bohr radius of hydrogen atom, then what is the de-Broglie wavelength (λ) of the electron present in the second orbit of hydrogen atom ? [n : any integer] **[JEE (Main) 2025]**

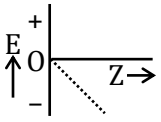
(1) $\frac{2a_0}{n\pi}$ (2) $\frac{8\pi a_0}{n}$ (3) $\frac{4\pi a_0}{n}$ (4) $\frac{4n}{\pi a_0}$

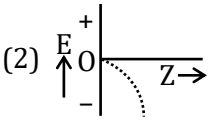
CAS436

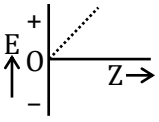
9. For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n ?

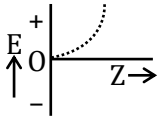
[E : Energy of the stationary state,

Z : atomic number, n = principal quantum number] **[JEE (Main) 2025]**

(1) 

(2) 

(3) 

(4) 

CAS437

10. The ionization energy of sodium in kJ mol^{-1} If electromagnetic radiation of wavelength 242 nm is just sufficient to ionize sodium atom is_____. **[JEE (Main) 2024]**

CAS333

11. The energy of an electron in the first Bohr orbit of hydrogen atom is -2.18×10^{-18} J. Its energy in the third Bohr orbit is_____. **[JEE (Main) 2023]**

(1) $\frac{1}{27}$ of this value (2) One third of this value

(3) Three times of this value (4) $\frac{1}{9}$ th of this value

Rankers Academy JEE

CAS182

12. The radius of the second Bohr orbit, in terms of the Bohr radius, a_0, in $Li^{2\oplus}$ is : **[JEE (Main) 2020]**

(1) $\frac{4a_0}{9}$ (2) $\frac{2a_0}{9}$ (3) $\frac{2a_0}{3}$ (4) $\frac{4a_0}{3}$

CAS046

13. The radius of the second Bohr orbit for hydrogen atom is :

(Planck's constant $h = 6.6262 \times 10^{-34}$ Js; mass of electron = 9.1091×10^{-31} kg; charge of electron $= 1.60210 \times 10^{-19}$ C ; permittivity of vaccum $\varepsilon_0 = 8.854185 \times 10^{-12}$ kg^{-1} m^{-3} A^2) **[JEE (Main) 2017]**

(1) 1.65Å (2) 4.76Å (3) 0.529Å (4) 2.12Å

CAS037

14. Energy of an electron is given by $E = -2.178 \times 10^{-18}$ J $\left(\frac{Z^2}{n^2}\right)$. Wavelength of light required to excite an electron in an hydrogen atom from level n = 1 to n = 2 will be :

(h = 6.62×10^{-34} Js and c = 3.0×10^8 ms^{-1}) **[JEE (Main) 2013]**

(1) 1.214×10^{-7} m (2) 2.816×10^{-7} m

(3) 6.500×10^{-7} m (4) 8.500×10^{-7} m

CAS034

For More Join: @IITJEE_Advanced

15. Heat treatment of muscular pain involves radiation of wavelength of about 900 nm. Which spectral line of H atom is suitable for this ?
Given: Rydberg constant
$R_H = 10^5\ cm^{-1}$, $h = 6.6 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m/s) **[JEE (Main) 2025]**

(1) Paschen series, $\infty \rightarrow 3$ (2) Lyman series, $\infty \rightarrow 1$
(3) Balmer series, $\infty \rightarrow 2$ (4) Paschen series, $5 \rightarrow 3$

CAS438

16. **Match List I with List II**

List I (Spectral Series for Hydrogen)		**List II (Spectral Region/Higher Energy State)**	
A.	Lyman	I.	Infrared region
B.	Balmer	II.	UV region
C.	Paschen	III.	Infrared region
D.	Pfund	IV.	Visible region

Choose the correct answer from the options given below :- **[JEE (Main) 2024]**

(1) A-II, B-III, C-I, D-IV (2) A-I, B-III, C-II, D-IV
(3) A-II, B-IV, C-III, D-I (4) A-I, B-II, C-III, D-IV

CAS335

17. The shortest wavelength of hydrogen atom in Lyman series is λ. The longest wavelength in Balmer series of He^+ is **[JEE (Main) 2023]**

(1) $\frac{5}{9\lambda}$ (2) $\frac{5}{9\lambda}$ (3) $\frac{36\lambda}{5}$ (4) $\frac{5\lambda}{9}$

Rankers Academy JEE

CAS179

18. For the Balmer series in the spectrum of H atom, $\bar{\nu} = R_H\left\{\frac{1}{n_1^2} - \frac{1}{n_2^2}\right\}$, the correct statements among (I) To (IV) are :
(I) As wavelength decreases, the lines in the series converge
(II) The integer n_1 is equal to 2
(III) The lines of longest wavelength corresponds to $n_2 = 3$
(IV) The ionization energy of hydrogen can be calculated from wave number of these lines

[JEE (Main) 2020]

(1) (II), (III), (IV) (2) (I), (II), (III)
(3) (I), (III), (IV) (4) (I), (II), (IV)

CAS045

19. For emission line of atomic hydrogen from $n_i = 8$ to $n_f = n$ the plot of wave number $(\bar{\nu})$ against $\left(\frac{1}{n^2}\right)$ will be (The Rydberg constant, R_H is in wave number unit). **[JEE (Main) 2019]**

(1) Linear with slope - R_H (2) Linear with intercept - R_H
(3) Non linear (4) Linear with slope R_H

CAS038

For More Join: @IITJEE_Advanced

20. Heat treatment of muscular pain involves radiation of wavelength of about 900 nm. Which spectral line of H-atom is suitable for this purpose ?
$[R_H = 1 \times 10^5\ cm^{-1}, h = 6.6 \times 10^{-34}\ Js, c = 3 \times 10^8\ ms^{-1}]$ **[JEE (Main) 2019]**

(1) Paschen, $5 \rightarrow 3$
(2) Paschen, $\infty \rightarrow 3$
(3) Lyman, $\infty \rightarrow 1$
(4) Balmer, $\infty \rightarrow 2$

CAS040

21. The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are: **[JEE (Main) 2019]**

(1) Paschen and Pfund
(2) Lyman and Paschen
(3) Brackett and Pfund
(4) Balmer and Brackett

CAS042

22. The limiting line in Balmer series of 'H' spectrum will have a frequency of:
(Rydberg constant, $R_H = 3.29 \times 10^{15}$ cycles/s) **[AIEEE 2012]**

(1) $3.65 \times 10^{14}\ s^{-1}$
(2) $8.22 \times 10^{14}\ s^{-1}$
(3) $3.29 \times 10^{15}\ s^{-1}$
(4) $5.26 \times 10^{13}\ s^{-1}$

CAS031

23. Given below are two statements:
Statement I: According to Bohr's model of hydrogen atom, the angular momentum of an electron in a given stationary state is quantised.
Statement II : The concept of electron in Bohr's orbit, violates the Heisenberg uncertainty principle.
In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE (Main) 2023]**

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are incorrect.

CAS184

24. The de Broglie wavelength of an electron in the 4th Bohr orbit is : **[JEE (Main) 2020]**

(1) $8\pi a_0$
(2) $2\pi a_0$
(3) $4\pi a_0$
(4) $6\pi a_0$

CAS048

25. If the kinetic energy of an electron is increased four times, the wavelength of the de-Broglie wave associated with it would become : **[AIEEE 2012]**

(1) Two times
(2) Half
(3) One fourth
(4) Four times

CAS032

26. If the radius of first orbit of H atom is a_0, the de-Broglie wavelength of an electron in the third orbit is : **[AIEEE 2012]**

(1) $6\ \pi\ a_0$
(2) $8\ \pi\ a_0$
(3) $2\ \pi\ a_0$
(4) $4\ \pi\ a_0$

CAS033

For More Join: @IITJEE_Advanced

27. Given below are two statements :

Statement (I): It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle.

Statement (II) : If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geq \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE (Main) 2025]

(1) **Statement I** is true but **Statement II** is false.
(2) Both **Statement I** and **Statement II** are true.
(3) **Statement I** is false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are false.

CAS439

28. The wave function (Ψ) of 2s is given by

$$\Psi_{2s} = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{1/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$$

At $r = r_0$, radial node is formed. Thus, r_0 in terms of a_0 **[JEE (Main) 2023]**

(1) $r_0 = a_0$ (2) $r_0 = 4a_0$ (3) $r_0 = \frac{a_0}{2}$ (4) $r_0 = 2a_0$

CAS181

29. The number of radial node/s for 3p orbital is: **[JEE (Main) 2024]**

(1) 1 (2) 4 (3) 2 (4) 3

CAS336

30. Arrange the following orbitals in decreasing order of energy ?

(A) n = 3, l = 0, m = 0 (B) n = 4, l = 0, m = 0
(C) n = 3, l = 1, m = 0 (D) n = 3, l = 2, m = 1

The correct option for the order is : **[JEE (Main) 2023]**

(1) B > D > C > A (2) D > B > C > A
(3) A > C > B > D (4) D > B > A > C

CAS362

31. The number of orbitals associated with quantum numbers n = 5, $s = +\frac{1}{2}$ is : **[JEE (Main) 2020]**

(1) 11 (2) 25 (3) 15 (4) 50

CAS044

32. The quantum number of four electrons are given below :

I. $n = 4, l = 2, m_\ell = -2, m_s = -\frac{1}{2}$ II. $n = 3, l = 2, m_\ell = 1, m_s = +\frac{1}{2}$

III. $n = 4, l = 1, m_\ell = 0, m_s = +\frac{1}{2}$ IV. $n = 3, l = 1, m_\ell = 1, m_s = -\frac{1}{2}$

The correct order of their increasing energies will be - **[JEE (Main) 2019]**

(1) IV < III < II < I (2) IV < II < III < I
(3) I < II < III < IV (4) I < III < II < IV

CAS041

For More Join: @IITJEE_Advanced

33. The total number of orbitals associated with the principal quantum number 5 is :

[JEE (Main) 2016]

(1) 25 (2) 5 (3) 20 (4) 10

CAS036

34. For hydrogen atom, the orbital/s with lowest energy is/are :

(A) 4s (B) $3p_x$ (C) $3d_{x^2-y^2}$ (D) $3d_{z^2}$ (E) $4p_z$

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) (A) and (E) only (2) (B) only

(3) (A) only (4) (B), (C) and (D) only

CAS440

35. Which of the following is/are not correct with respect to energy of atomic orbitals of hydrogen atom?

(A) 1s < 2p < 3d < 4s (B) 1s < 2s = 2p < 3s = 3p

(C) 1s < 2s < 2p < 3s < 3p (D) 1s < 2s < 4s < 3d

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) (B) and (D) only (2) (A) and (C) only

(3) (C) and (D) only (4) (A) and (B) only

CAS441

36. Given below are two statements:

Statement-I: The orbitals having same energy are called as degenerate orbitals.

Statement-II: In hydrogen atom, 3p and 3d orbitals are not degenerate orbitals.

In the light of the above statements, choose the **most appropriate** answer from the options given **[JEE (Main) 2024]**

Pankers Academy JEE

(1) Statement-I is true but Statement-II is false

(2) Both Statement-I and Statement-II are true.

(3) Both Statement-I and Statement-II are false

(4) Statement-I is false but Statement-II is true

CAS337

37. The 71st electron of an element X with an atomic number of 71 enters into the orbital :

[JEE (Main) 2019]

(1) 4f (2) 6p (3) 6s (4) 5d

CAS039

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Rutherford's experiment, which established the nuclear model of atom, used a beam of :–

[JEE (Advanced) 2002]

(A) β-particles, which impinged on a metal foil and get absorbed.
(B) γ-rays, which impinged on a metal foil and ejected electron.
(C) Helium atoms, which impinged on a metal foil and got scattered.
(D) Helium nuclei, which impinged on a metal foil and got scattered.

CAS071

2. Consider a helium (He) atom that absorbs a photon of wavelength 330 nm. The change in the velocity (in cm s^{-1}) of He atom after the photon absorption is ___.
(Assume: Momentum is conserved when photon is absorbed.
Use: Planck constant = 6.6×10^{-34} J s, Avogadro number = 6×10^{23} mol^{-1}, Molar mass of He = 4 g mol^{-1})

[JEE (Advanced) 2021]

CAS093

3. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is :-

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ(eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

[JEE (Advanced) 2011]

CAS083

4. According to Bohr's model, the highest kinetic energy is associated with the electron in the :

[JEE (Advanced) 2024]

(A) first orbit of H atom
(B) first orbit of He^+
(C) second orbit of He^+
(D) second orbit of Li^{2+}

CAS338

5. For $He^{\oplus}$, a transition takes place from the orbit of radius 105.8 pm to the orbit of radius 26.45 pm. The wavelength (in nm) of the emitted photon during the transition is ___.
[**Use:**
Bohr radius, a = 52.9 pm
Rydberg constant, $R_H = 2.2 \times 10^{-18}$ J
Planck's constant, $h = 6.6 \times 10^{-34}$ J s
Speed of light, $c = 3 \times 10^8$ m s^{-1}]

[JEE (Advanced) 2023]

CAS185

Answer the following by appropriately matching the lists based on the information given in the paragraph : (Q. 6-7)
Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following List-I contains some quantities for the nth orbit of the atom and List-II contains options showing how they depend on n.

List-I	**List-II**
(I) Radius of the nth orbit	(P) $\propto n^{-2}$
(II) Angular momentum of the electron in the nth orbit	(Q) $\propto n^{-1}$
(III) Kinetic energy of the electron in the nth orbit	(R) $\propto n_0$
(IV) Potential energy of the electron in the nth orbit	(S) $\propto n^1$
	(T) $\propto n^2$
	(U) $\propto n^{1/2}$

For More Join: @IITJEE_Advanced

6. Which of the following options has the correct combination considering List-I and List-II?

[JEE (Advanced) 2019]

(A) (II), (R) (B) (I), (P)
(C) (I), (T) (D) (II), (Q)

CAS090

7. Which of the following options has the correct combination considering List-I and List-II?

[JEE (Advanced) 2019]

(A) (III), (S) (B) (IV), (Q)
(C) (IV), (U) (D) (III), (P)

CAS091

8. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius]

[JEE (Advanced) 2012]

(A) $\frac{h^2}{4\pi^2 m a_0^2}$ (B) $\frac{h^2}{16\pi^2 m a_0^2}$

(C) $\frac{h^2}{32\pi^2 m a_0^2}$ (D) $\frac{h^2}{32\pi^2 m a_0^2}$

CAS084

9. Given in hydrogen atom r_n, V_n, E, K_n stand for radius, potential energy, total energy and kinetic energy in n^{th} orbit. Find the value of u, v, x, y.

(1) $U = \frac{V_n}{K_n}$ (P) 1

(2) $\frac{1}{r_n} \propto E^x$ (Q) –2

(3) $r_n \propto Z^y$ (Z = Atomic number) (R) –1

(4) v = (Orbital angular momentum of electron in its lowest energy) (S) 0

[JEE (Advanced) 2006]

(A) (1) S, (2) P, (3) R, (4) Q (B) (1) Q, (2) S, (3) P, (4) R
(C) (1) P, (2) R, (3) S, (4) Q (D) (1) Q, (2) P, (3) R, (4) S

CAS077

10. The velocity of electron in first Bohr orbit of hydrogen atom is **[JEE (Advanced) 2005]**

(A) 2.197×10^6 m/s (B) 2.197×10^8 m/s
(C) 2.197×10^4 m/s (D) 4.197×10^6 m/s

CAS074

11. The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom?

[JEE (Advanced) 2004]

(A) $He^{\oplus}$ (n = 2) (B) $Li^{2\oplus}$ (n = 2)
(C) $Li^{2\oplus}$ (n = 3) (D) $Be^{3\oplus}$(n = 2)

CAS072

12. The De-Broglie wavelength of the electron in first Bohr orbit is (Given $r = a_0$)

[JEE (Advanced) 2005]

(A)3.8 Å (B) 3.32 Å (C) 4.8 Å (D) 2.4 Å

CAS075

Rankers Academy JEE

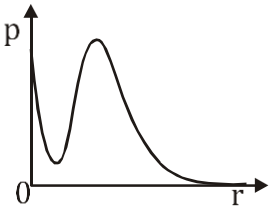
For More Join: @IITJEE_Advanced

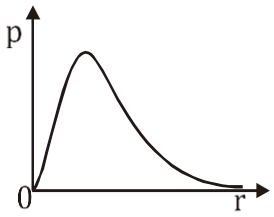
13. Among the following, the correct statement(s) for electrons in an atom is(are)
[JEE (Advanced) 2024]

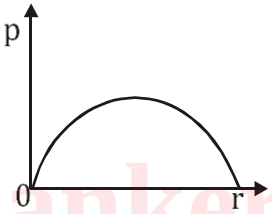
(A) Uncertainty principle rules out the existence of definite paths for electrons.
(B) The energy of an electron in 2*s* orbital of an atom is lower than the energy of an electron that is infinitely far away from the nucleus.
(C) According to Bohr's model, the most negative energy value for an electron is given by n = 1, which corresponds to the most stable orbit.
(D) According to Bohr's model, the magnitude of velocity of electrons increases with increase in values of n.

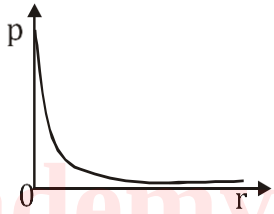
CAS339

14. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr, at a distance r from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is -
[JEE (Advanced) 2016]

(A) 

(B) 

(C) 

(D) 

Rankers Academy JEE

CAS085

Paragraph for questions 15 to 17

The hydrogen-like species $Li^{2\oplus}$ is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

15. The state S_1 is :-
[JEE (Advanced) 2010]

(A) 1s (B) 2s (C) 2p (D) 3s

CAS079

16. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-
[JEE (Advanced) 2010]

(A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50

CAS080

17. The orbital angular momentum quantum number of the state S_2 is :-
[JEE (Advanced) 2010]

(A) 0 (B) 1 (C) 2 (D) 3

CAS081

18. The Schrodinger wave equation for hydrogen atom is

$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r_0}{a_0}\right)e^{-r/a}$$

Where a_0 is Bohr's radius. Let the radial node in 2*s* be at r_0. Then correct value of r_0 in terms of a_0 is
[JEE (Advanced) 2004]

(A) $2a_0$ (B) a_0 (C) $4a_0$ (D) $a_0/2$

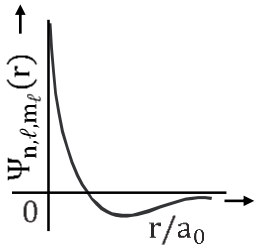
CAS073

For More Join: @IITJEE_Advanced

TABLE TYPE QUESTION

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

The wave function ψ_{n,ℓ,m_ℓ} is a mathematical function whose value depends upon spherical polar coordinates (r, θ, ϕ) of the electron and characterized by the quantum numbers n, ℓ and m_ℓ. Here r is distance from nucleus, θ is colatitude and ϕ is azimuth. In the mathematical functions given in the Table, Z is atomic number a_0 is Bohr radius.

Column-1	Column-2	Column-3
(I) 1s orbital	(i) $\psi_{n,\ell,m_\ell} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_e}\right)}$	(P) 
(II) 2s orbital	(ii) One radial node	(Q) Probability density at nucleus $\propto \frac{1}{a_0^3}$
(III) $2p_z$ orbital	(iii) $\psi_{n,\ell,m_\ell} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{2a_0}\right)} \cos\theta$	(R) Probability density is maximum at nucleus
(IV) $3d_{z^2}$ orbital	(iv) xy - plane is a nodal plane	(S) Energy needed to excite electron from n = 2 state to n = 4 state is $\frac{27}{32}$ times the energy needed to excite electron from n = 2 state to n=6 state

Rankers Academy JEE

19. For the given orbital in column 1, the only CORRECT combination for any hydrogen - like species is : **[JEE (Advanced) 2017]**

(A) (IV) (iv) (R) (B) (III) (iii) (P)
(C) (I) (ii) (S) (D) (II) (ii) (P)

CAS086

20. For $He^{\oplus}$ ion, the only INCORRECT combination is **[JEE (Advanced) 2017]**

(A) (I) (iii) (R) (B) (II) (ii) (Q)
(C) (I) (i) (R) (D) (I) (i) (S)

CAS087

21. For hydrogen atom, the only CORRECT combination is **[JEE (Advanced) 2017]**

(A) (I) (iv) (R) (B) (II) (i) (Q)
(C) (I) (i) (S) (D) (I) (i) (P)

CAS088

22. The orbital angular momentum of 2p-orbital is (in terms of $h/2\pi$) **[JEE (Advanced) 2005]**

(A) $3\sqrt{2}.\frac{h}{2\pi}$ (B) $2\sqrt{2}.\frac{h}{2\pi}$ (C) $\sqrt{2}.\frac{h}{2\pi}$ (D) $\sqrt{3}.\frac{h}{2\pi}$

CAS076

For More Join: @IITJEE_Advanced

23. The ground state energy of hydrogen atom is –13.6 eV. Consider an electronic state Ψ of $He^{\oplus}$ whose energy, azimuthal quantum number and magnetic quantum number are –3.4 eV, 2 and 0 respectively. Which of the following statement(s) is(are) true for the state Ψ ?

[JEE (Advanced) 2019]

(A) It has 2 angular nodes
(B) It has 3 radial nodes
(C) It is a 4d state
(D) The nuclear charge experienced by the electron in this state is less than 2e, where e is the magnitude of the electronic charge.

CAS089

24. In an atom, the total number of electrons having quantum numbers n=4, $|m_l| = 1$ and $m_s = \frac{1}{2}$ –is:

[JEE (Advanced) 2014]

CAS363

25. Not considering the electronic spin the degeneracy of the second excited state (n = 3) of H-atom is 9, where the degeneracy of the second excited state of H^- is **[JEE (Advanced) 2015]**

CAS364

26. The maximum number of electrons that can have principal quantum number, n = 3, and spin quantum number, s = – 1/2, is **[JEE (Advanced) 2011]**

CAS082

27. Match the entries in Column I with the correctly related quantum number(s) in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

Rankers Academy JEE

Column I		Column II
(A) Orbital angular momentum of the electron in a hydrogen-like atomic orbital		(P) Principal quantum number
(B) A hydrogen-like one-electron wave function obeying Pauli principle		(Q) Azimuthal quantum number
(C) Shape, size and orientation of hydrogen like atomic orbitals		(R) Magnetic quantum number
(D) Probability density of electron at the nucleus in hydrogen-like atom		(S) Electron spin quantum number

[JEE (Advanced) 2008]

(A) (A) Q, R ; (B) P, Q, R, S ; (C) P, Q, R ; (D) P, Q
(B) (A) R, Q ; (B) P, Q, R ; (C) P, Q, R ; (D) P, Q
(C) (A) R, Q ; (B) P, Q, R, S ; (C) Q, R ; (D) P, Q
(D) (A) R, S ; (B) P, Q, S ; (C) P, Q, R ; (D) P, Q

CAS078

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	C	B	C	D	D	D	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	A	C	C	D	A	B	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	D	D	C	A	B	C	D	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	A	B	B	A	C	C	A	D	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	B	A	C	A	D	B	B	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	C	A	B	A	C	B	C	C	A
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	A	C	B	B	A	C	B	A	B	C
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	A	B	D	D	C	B	C	D	A	C
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	A	C	C	B	A	C	A	A	A	B
Que.	91	92	93	94						
Ans.	C	A	C	B						

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AC	BCD	CD	BD	BC	ABC	ABCD	BC	BD	BD
Que.	11	12	13				14			
Ans.	ABD	BCD	(A) S, (B) R (C) Q (D) P				(A) P, (B) P,Q,S (C) P,R (D) Q,S			
Que.	15	16	17	18	19	20	21	22	23	24
Ans.	D	C	C	B	B	A	C	B	C	A
Que.	25	26	27	28						
Ans.	C	A	C	A						

Aakash Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (S)

1.	1	**2.**	1	**3.**	8	**4.**	4	**5.**	150
6.	1	**7.**	3	**8.**	50	**9.**	2	**10.**	1
11.	9	**12.**	2	**13.**	4	**14.**	65	**15.**	2
16.	5280	**17.**	3	**18.**	90	**19.**	3	**20.**	8
21.	8	**22.**	5	**23.**	6	**24.**	2	**25.**	3
26.	9	**27.**	8	**28.**	0	**29.**	2	**30.**	6
31.	1.60	**32.**	1.28	**33.**	0.50	**34.**	1.44	**35.**	0.20
36.	0.62	**37.**	7.50	**38.**	2.66	**39.**	0.50	**40.**	3.06
41.	3.06	**42.**	(a) 75.00; (b) 1.65			**43.**	0.26		
44.	5.40 to 5.50	**45.**	8.20 to 8.40	**46.**	0.85	**47.**	6566.40; 1216.00; 1026.00		

48. $E = \frac{n^6h^6}{384 \times \pi^6 . m^3 \cdot k^2 \cdot e^4}$ **49.** 435.20 **50.** 9.68 or 9.69

51.	972.54 or 972.55			**52.**	300.00, 0.26	**53.**	6.03	**54.**	0.11
55.	0.17	**56.**	0.50	**57.**	3.75	**58.**	0.63		

59. (i) 2.00; (ii) 9.60 to 9.80 **60.** (a) Z = 3 ; (b) $n_1 = 12$, $n_2 = 15$; (c) 0.08

EXERCISE - JEE (Main) PYQ

Bankers Academy JEE

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	798	2	3	4	10	4	2	2	494
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	4	1	1	3	2	2	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	1	1	2	1	2	4	1	2
Que.	31	32	33	34	35	36	37			
Ans.	2	2	1	4	3	1	4			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	30	4	B	30	C	D	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	ABC	B	B	C	B	A	D	A
Que.	21	22	23	24	25	26	27			
Ans.	C	C	AC	6	3	9	A			

For More Join: @IITJEE_Advanced

Important Notes

Rankers Academy JEE

For More Join: @IITJEE_Advanced